

Online Fair Division with Subsidy: When Do Envy-Free Allocations Exist, and at What Cost?

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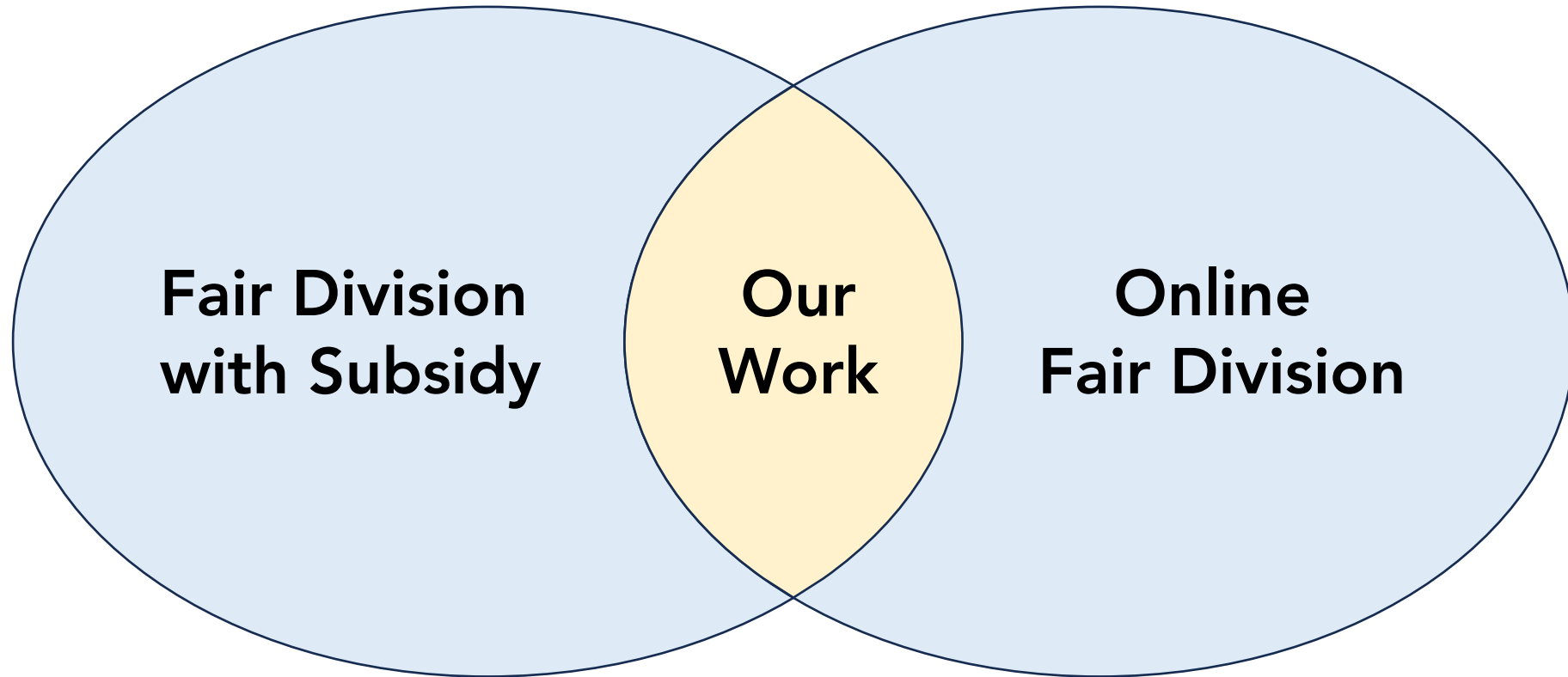
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Online Fair Division with Subsidy



Fair Division of Indivisible Goods

Fairly divide m indivisible goods among n agents.

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For now, assume **additive** valuations,
where agent i 's value for bundle X_i is $v_i(X_i) = \sum_{g \in X_i} v_{i,g}$

Envy-Freeness

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Envy-free allocations might **not** exist.

Fair Division with Subsidy

Allocation X with **subsidy vector** $p = (p_1, \dots, p_n)$ is **envy-free** if $v_i(X_i) + p_i \geq v_i(X_j) + p_j$ for all i, j .

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	item a	item b	item c	subsidy
agent 1	0.75	0.75	1	0
agent 2	1	1	0.5	0.5

$$v_2(X_2) + p_2 = 1.5 \quad v_2(X_1) + p_1 = 1.5 \quad \Rightarrow \quad \text{no envy}$$

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This allocation is **envy-free** with subsidy!

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Very active area of research.
Focus on minimizing **total subsidy** $\sum_{i \in [n]} p_i$.

[Maskin, 1987] [Tadenuma, Thomson, 1993] [Aragones, 1995] [Klijn, 2000]
[Halpern, Shah, SAGT 2019] [Brustle, Dippel, Narayan, Suzuki, Vetta, EC 2020]
[Barman, Krishna, Narahari, Sadhukhan, IJCAI 2022] [Caragiannis, Ioannidis, WINE 2022]
[Goko, Igarashi, Kawase, Makino, Sumita, Tamura, Yokoi, Yokoo, AAMAS 2022]
[Choo, Ling, Suksompong, Teh, Zhang, ORL 2024] [Wu, Zhang, Zhou, WINE 2024]
[Dupre la Tour, Fujii, 2025] [Aziz, Huang, Kimura, Saha, Sun, Suzuki, Yokoo, AAMAS 2025]

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When does X admit p making it **envy-free**?

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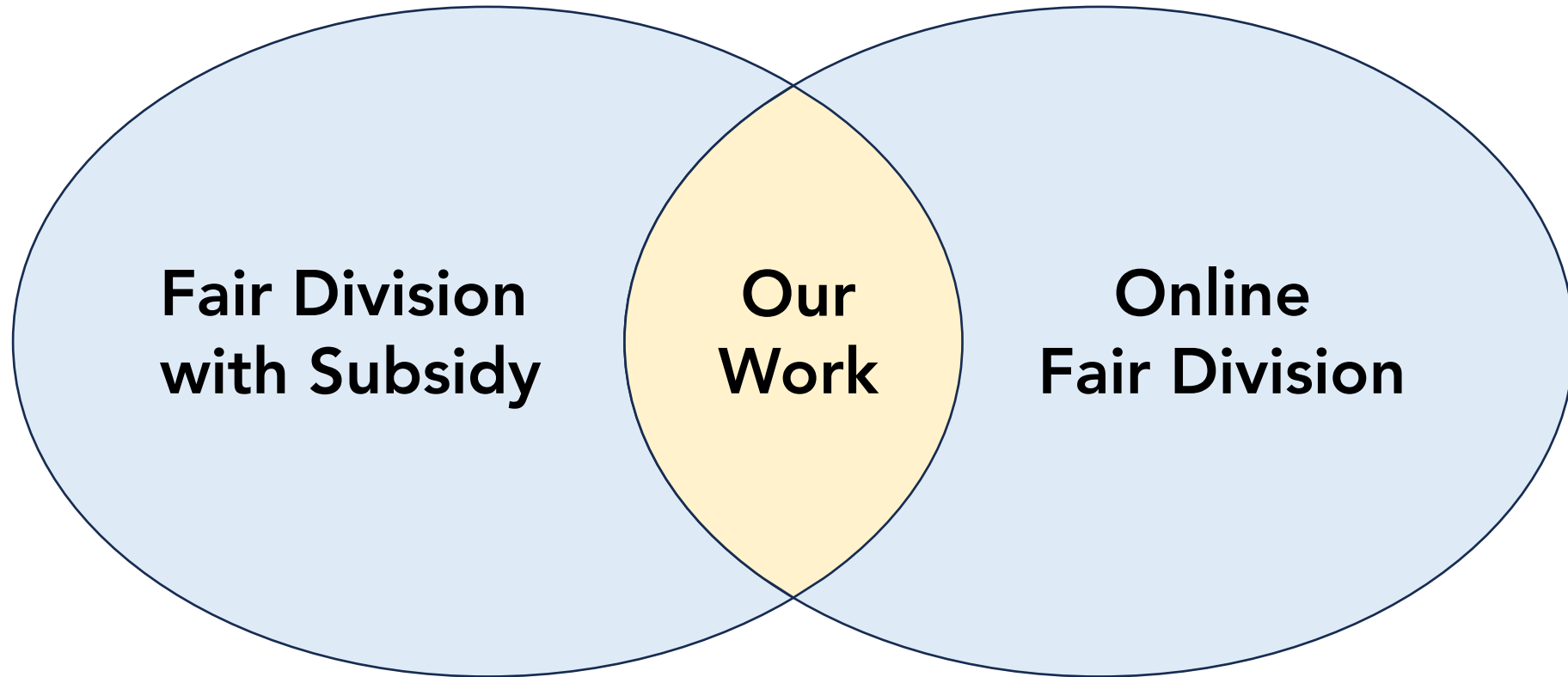
When does X admit p making it **envy-free**?

Theorem: X is **envy-freeable** iff X is **locally efficient**: there exists *no* permutation $\pi : [n] \rightarrow [n]$ with

$$\sum_{i \in [n]} v_i(X_{\pi(i)}) > \sum_{i \in [n]} v_i(X_i).$$

[Aragones, 1995] [Haake, Raith, Su, 2002] [Halpern, Shah, SAGT 2019]
[Hartline, leong, Mu'alem, Schapira, Zohar, ADT 2009]

Online Fair Division with Subsidy



Online Fair Division

Items are arriving **online**, while agents are **offline**.

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new item arrives



	item a	item b	item c	item d
agent 1	0.75	0.75	1	1
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Items are arriving **online**, while agents are **offline**.

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We **can't modify** past allocations.
Future items are **unknown**.

Online Fair Division

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Very active area of research.
Strong **impossibilities**: e.g., cannot achieve
envy-freeness up to some good (EF1).

[Aleksandrov, Aziz, Gaspers, Walsh, IJCAI 2015]

[He, Procaccia, Psomas, Zeng, IJCAI 2019]

[Gkatzelis, Psomas, Tan, AAAI 2021] [Zhou, Bai, Wu, ICML 2023]

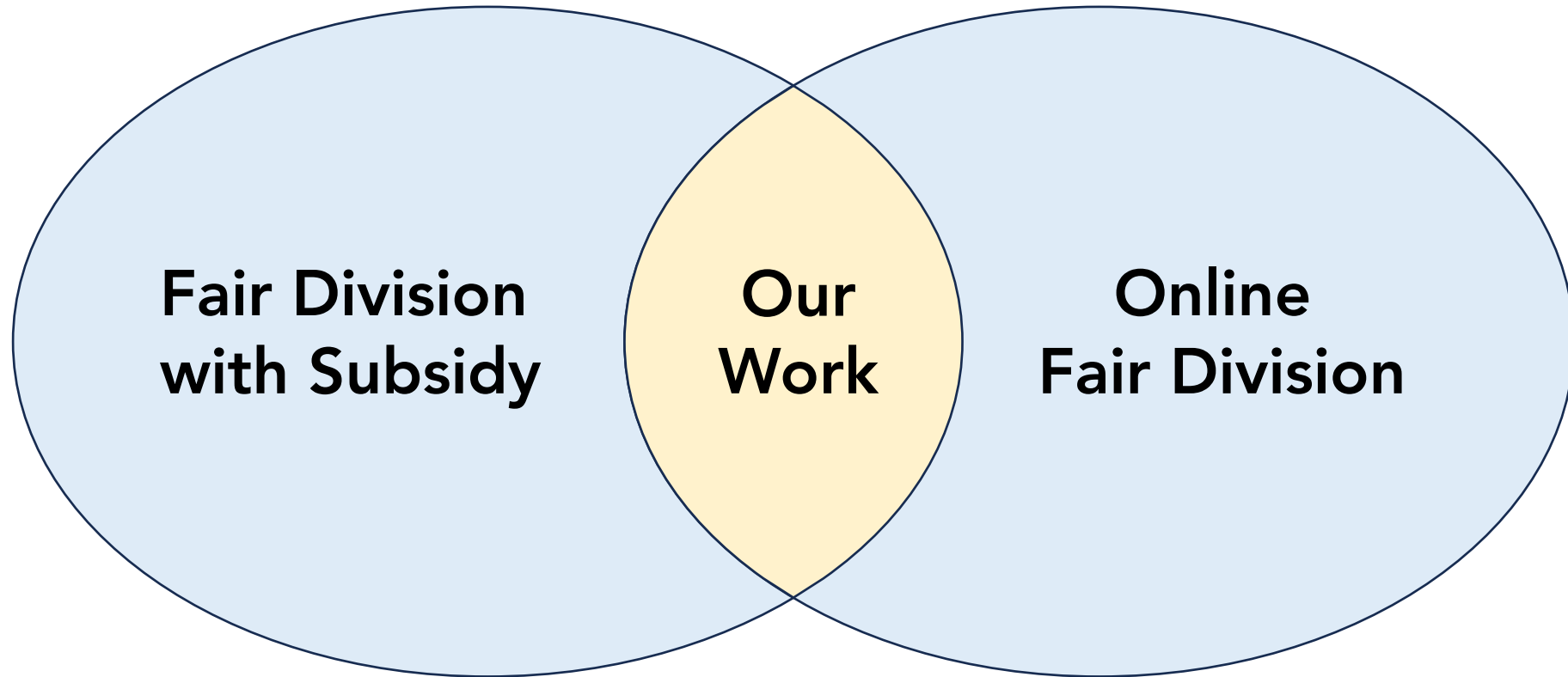
[Benade, Kazachkov, Procaccia, Psomas, Zeng, OR 2024]

[Halpern, Psomas, Verma, Xie, EC 2025] [Neoh, Peters, Teh, 2025]

[Amanatidis, Lolos, Markakis, Turme, SAGT 2025] [Wang, Wei, AAAI 2026]

[Choo, Fu, Khu, Neoh, Poon, Teh, 2025]

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Starting Point

Theorem: For **additive** valuations, there is an **online** algorithm that is **envy-free** with **total subsidy** $m(n - 1)$.

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Proof: Assign each item to agent who values it most.
maximizes welfare $\Rightarrow$ **local efficiency** $\Rightarrow$ **envy-freeability**

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maximizes welfare $\Rightarrow$ **local efficiency** $\Rightarrow$ **envy-freeability**

	item 1	...	item m	subsidy	} total subsidy $m(n - 1)$
agent 1	1	1	1	0	
agent 2	$1 - \epsilon$	$1 - \epsilon$	$1 - \epsilon$	$m(1 - \epsilon)$	
$\vdots$	$1 - \epsilon$	$1 - \epsilon$	$1 - \epsilon$	$m(1 - \epsilon)$	
agent n	$1 - \epsilon$	$1 - \epsilon$	$1 - \epsilon$	$m(1 - \epsilon)$	

Research Direction

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Question 1: Can we go beyond **additive** valuations?

Question 2: Is it possible to use **small subsidy** that does not grow with the number of items m ?

Q1: Beyond Additive Valuations

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	Can EF be achieved online? (Our Work)
Additive	YES
SPLC	YES
Binary Submodular	NO
Binary Supermodular	NO
Budget Additive	NO
General Monotone	NO

Q1: Beyond Additive Valuations

	Can EF be achieved online? (Our Work)	Can EF be achieved offline? (Prior Work)
Additive	YES	YES
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Strong **online vs. offline** separation. Three **incomparable** impossibility results for **highly structured** classes.

Q2: Small Subsidy

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Online EF with
small subsidy?

Additive

NO

k -Demand

YES

k -Valued

YES

Rank-One

YES

Restricted Additive

YES

Identical Monotone

YES

Q2: Small Subsidy

	Online EF with small subsidy?	Online subsidy bound
Additive	NO	$m(n - 1)$
k -Demand	YES	$k(n - 1)$
k -Valued	YES	$\leq n^2 k^n$
Rank-One	YES	$n(n + 1)/2 - 1$
Restricted Additive	YES	$n(n - 1)/2$
Identical Monotone	YES	$n - 1$

Q2: Small Subsidy

	Online EF with small subsidy?	Online subsidy bound	Offline subsidy bound (Prior Work)
Additive	NO	$m(n - 1)$	$n - 1$
k -Demand	YES	$k(n - 1)$	$n - 1$
k -Valued	YES	$\leq n^2 k^n$	$n - 1$
Rank-One	YES	$n(n + 1)/2 - 1$	$n - 1$
Restricted Additive	YES	$n(n - 1)/2$	$n - 1$
Identical Monotone	YES	$n - 1$	$n - 1$

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Rank-One	YES	$n(n + 1)/2 - 1$	$n - 1$
Restricted Additive	YES	$n(n - 1)/2$	$n - 1$
Identical Monotone	YES	$n - 1$	$n - 1$

Subsidy bounds are **tight** (except k -valued). **Offline** subsidy for additive is $n - 1$; **online** is $\Omega(n^2)$ even for **structured** subclasses.

Q2: Small Subsidy

Thm: There is *no* **online** algorithm that achieves **envy-freeness** with **total subsidy** less than $m(n - 1)$ for **additive** valuations.

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	item 1	item 2	item 3	...	subsidy
agent 1	$1 - \epsilon + 2\delta$	$1 - \epsilon + 4\delta$	$1 - \epsilon + 8\delta$	...	0
agent 2	$1 - \epsilon + \delta$	$1 - \epsilon + 2\delta$	$1 - \epsilon + 4\delta$	...	$\approx m$
$\vdots$	$1 - \epsilon + \delta$	$1 - \epsilon + 2\delta$	$1 - \epsilon + 4\delta$	...	$\approx m$
agent n	$1 - \epsilon + \delta$	$1 - \epsilon + 2\delta$	$1 - \epsilon + 4\delta$	...	$\approx m$

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agent n	$1 - \epsilon + \delta$	$1 - \epsilon + 2\delta$	$1 - \epsilon + 4\delta$	...	$\approx m$

Local efficiency at every step forces **all items** to agent 1.

Connection to Envy Minimization

We focus on achieving **envy-freeness** with **small subsidy**.

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Related line of work studies **online envy minimization**:
maintain allocation with $v_i(X_j) - v_i(X_i) \leq E$ for small E .

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Lemma: Any online algorithm that maintains **local efficiency** and has **envy bounded** by E gives **envy-freeness** with total **subsidy** at most $E \cdot n(n - 1)$.

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Lemma: Any online algorithm that maintains **local efficiency** and has **envy bounded** by E gives **envy-freeness** with total **subsidy** at most $E \cdot n(n - 1)$.

Local efficiency is the source of hardness in our setting.

Summary

Question 1: Is it possible to maintain **envy-freeability online** beyond **additive** valuations?

Additive	YES
SPLC	YES
Binary Submodular	NO
Binary Supermodular	NO
Budget Additive	NO

Question 2: Is it possible to achieve **envy-freeness online** using **small subsidy** that does not grow with m ?

Additive	NO	$m(n - 1)$
k -Demand	YES	$k(n - 1)$
k -Valued	YES	$\leq n^2 k^n (?)$
Rank-One	YES	$n(n + 1)/2 - 1$
Restricted Additive	YES	$n(n - 1)/2$
Identical Monotone	YES	$n - 1$